

Microprocessors and Microcontrollers Lab

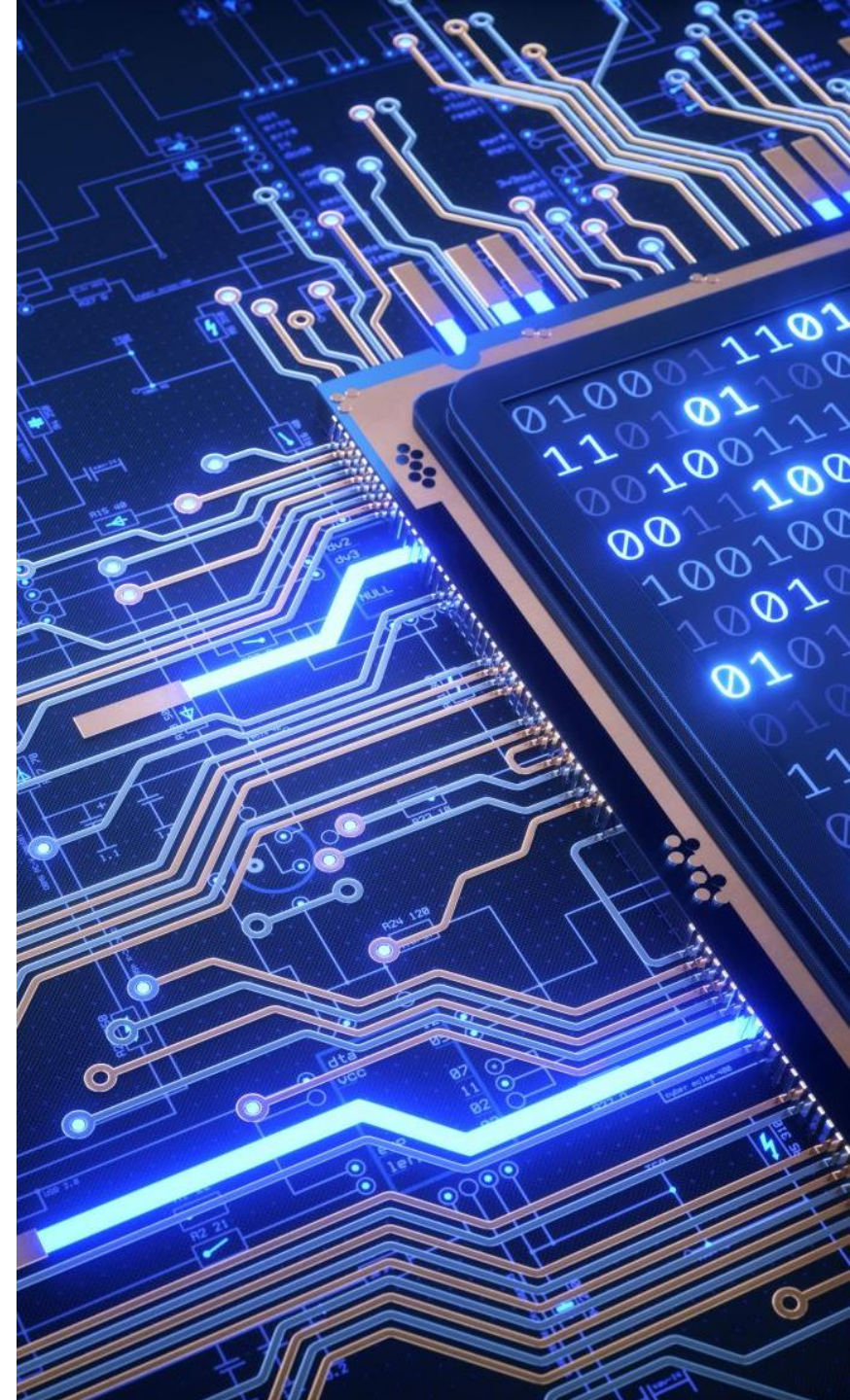
BECE204P

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Course Objectives

1. To familiarize the students with assembly language programming using microprocessor and microcontroller.
2. To interface peripherals and I/O devices with the microcontroller and microprocessor.



Course Outcome

□ Student will be able to

1. Showcase the skill, knowledge and ability of programming microcontroller and microprocessor using its instruction set.
2. Expertise with microcontroller and interfaces including general purpose input/ output, timers, serial communication, LCD, keypad and ADC.

List of Experiments

S.NO.	Experiment
1	Arithmetic Operation
2	Data Transfer Between ROM and RAM
3	Stack Operation
4	Digital Circuit Implementation
5	Port Programming
6	Timer and Counter
7	Serial Communication
8	Interrupts

Assessment Rubrics

Mode of Assessment	Marks
1. Internal Assessment-1	12 Marks
2. Internal Assessment-2	12 Marks
3. Internal Assessment-3	12 Marks
4. Internal Assessment-4	12 Marks
5. Internal Assessment-5	12 Marks

Marks Distribution for all Internal Assessment based on your Class Performance (50%) + VTOP Documentations (50%).

6. Final Assessment (FAT) 40% (FAT Exam will be conducted for 50 Marks)

Contact Details

- ❑ Cabin: Building: **SJT Annex, Room No. 103C**
- ❑ Email: vishal.gupta@vit.ac.in
- ❑ Contact: **+91-8965044467 (WhatsApp Only)**

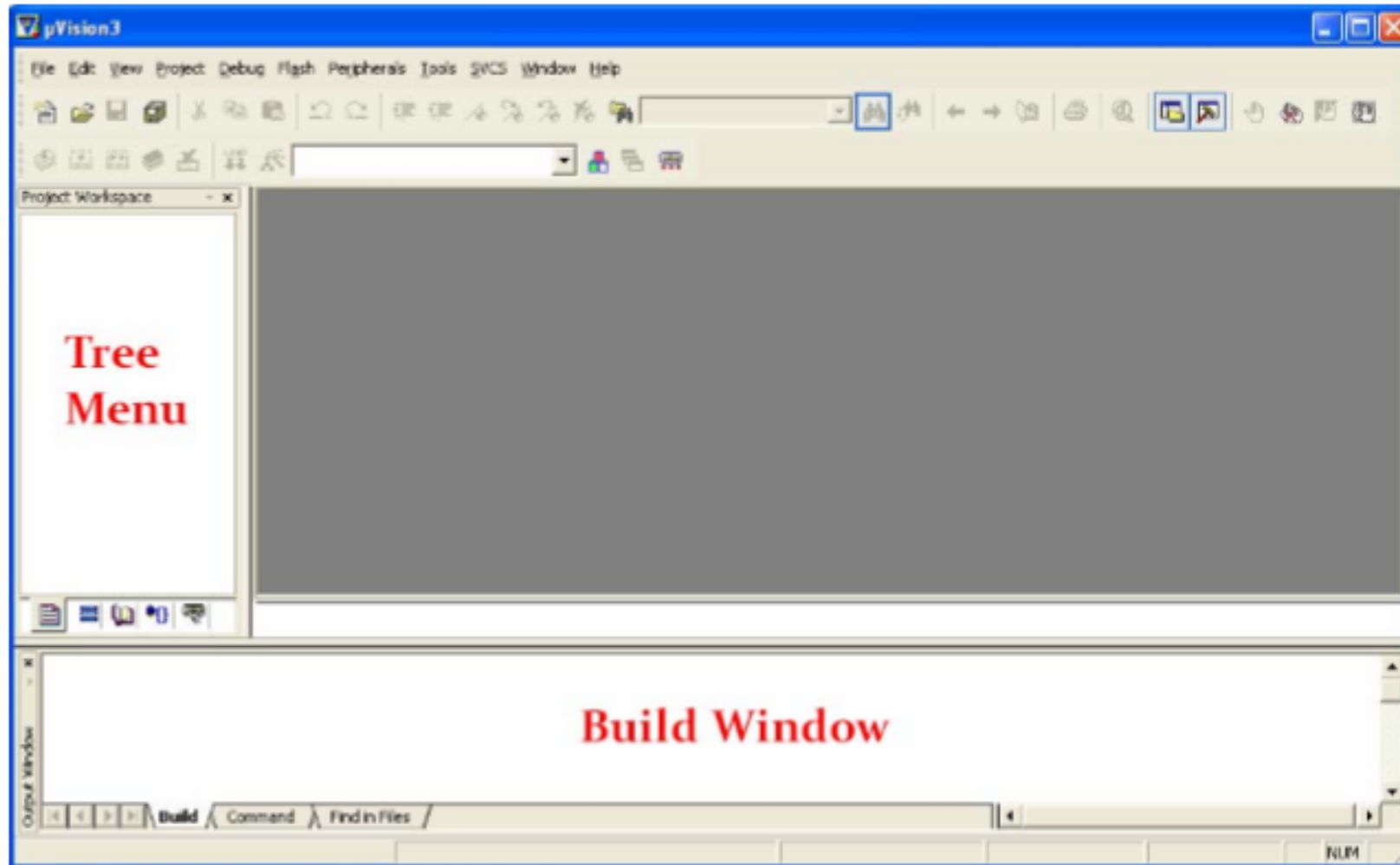


Instructions to use KEIL Simulator

https://drive.google.com/file/d/1m4vamxnywVRohnFoZFp5jwzjF1_hr6fQ/view

Or you can also download from the Keil official website

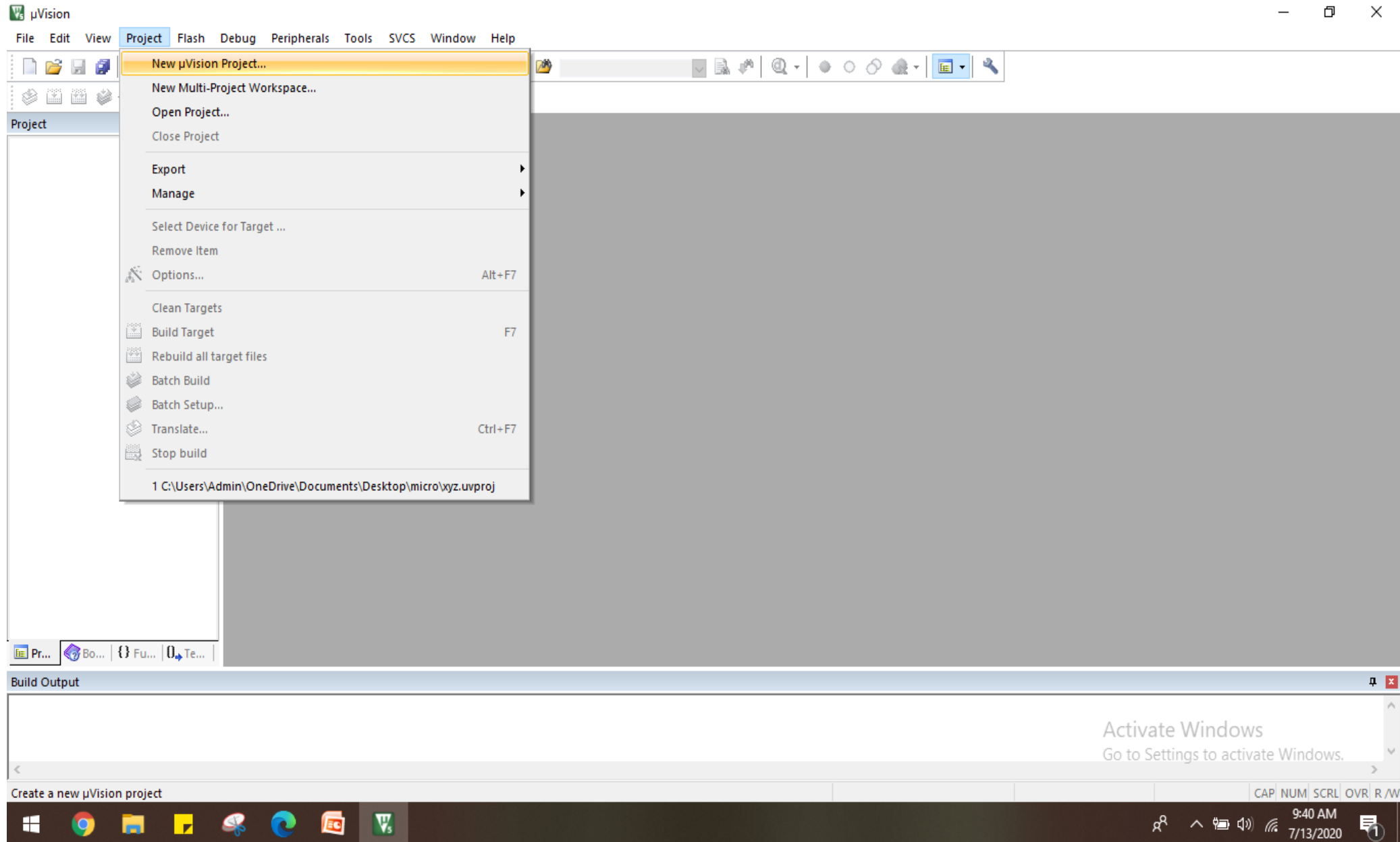
<https://www.keil.com/demo/eval/c51.htm>



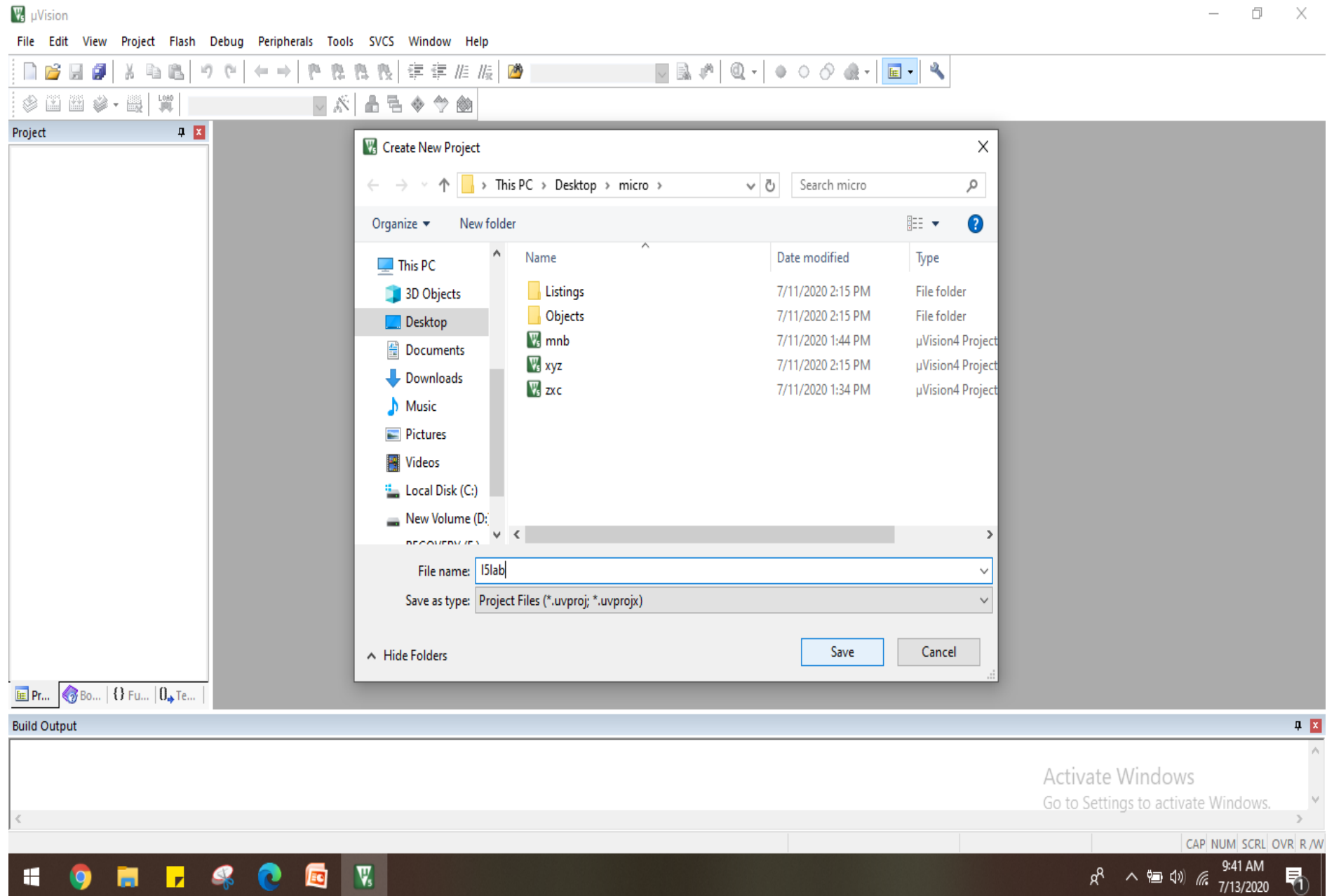
STEP 1: Open Keil uVision5

**(Start > Programs> Keil
uVision 5)**

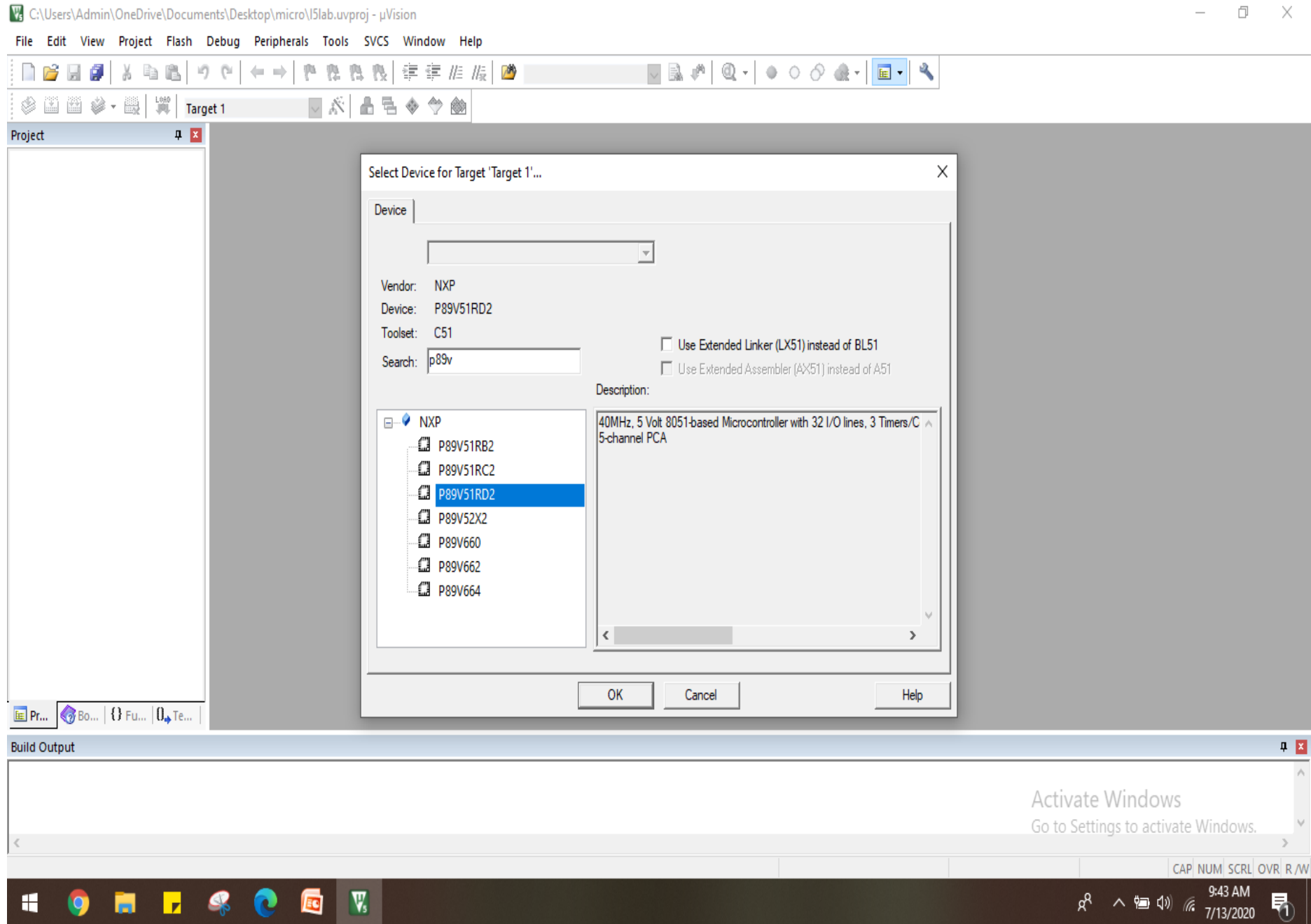
STEP2: Start a new project (File>New>µvision project)



STEP3: Type any name without extension, and Save in your folder



STEP 4: Select Nxp >P89V51RD2



The screenshot shows the µVision IDE interface. A dialog box titled "Select Device for Target 'Target 1'..." is open. The "Device" field is empty. The "Vendor" is set to "NXP", the "Device" is "P89V51RD2", and the "Toolset" is "C51". The "Search" field contains "p89v". The "Description" field shows "40MHz, 5 Volt 8051-based Microcontroller with 32 I/O lines, 3 Timers/C 5-channel PCA". The "OK" button is highlighted. The background shows the µVision IDE with a "Project" window on the left and a "Build Output" window at the bottom. The Windows taskbar is visible at the bottom of the screen.

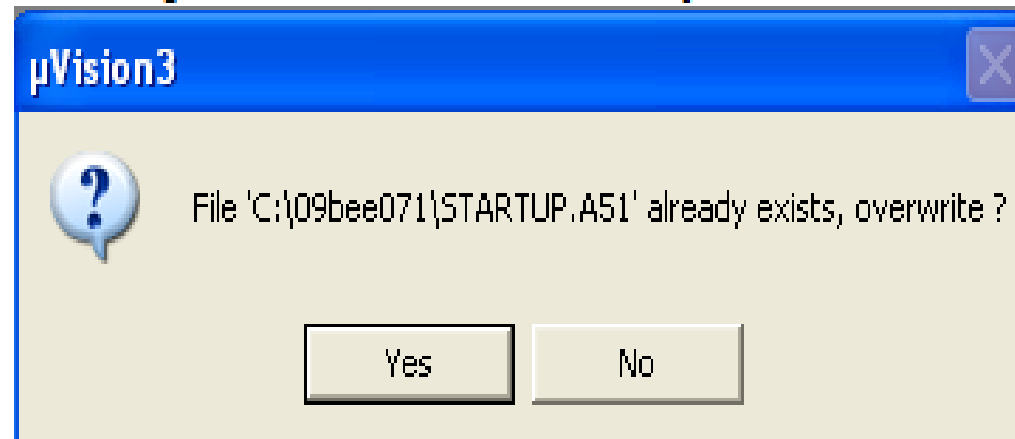
Activate Windows
Go to Settings to activate Windows.

STEP 5 (select→yes)

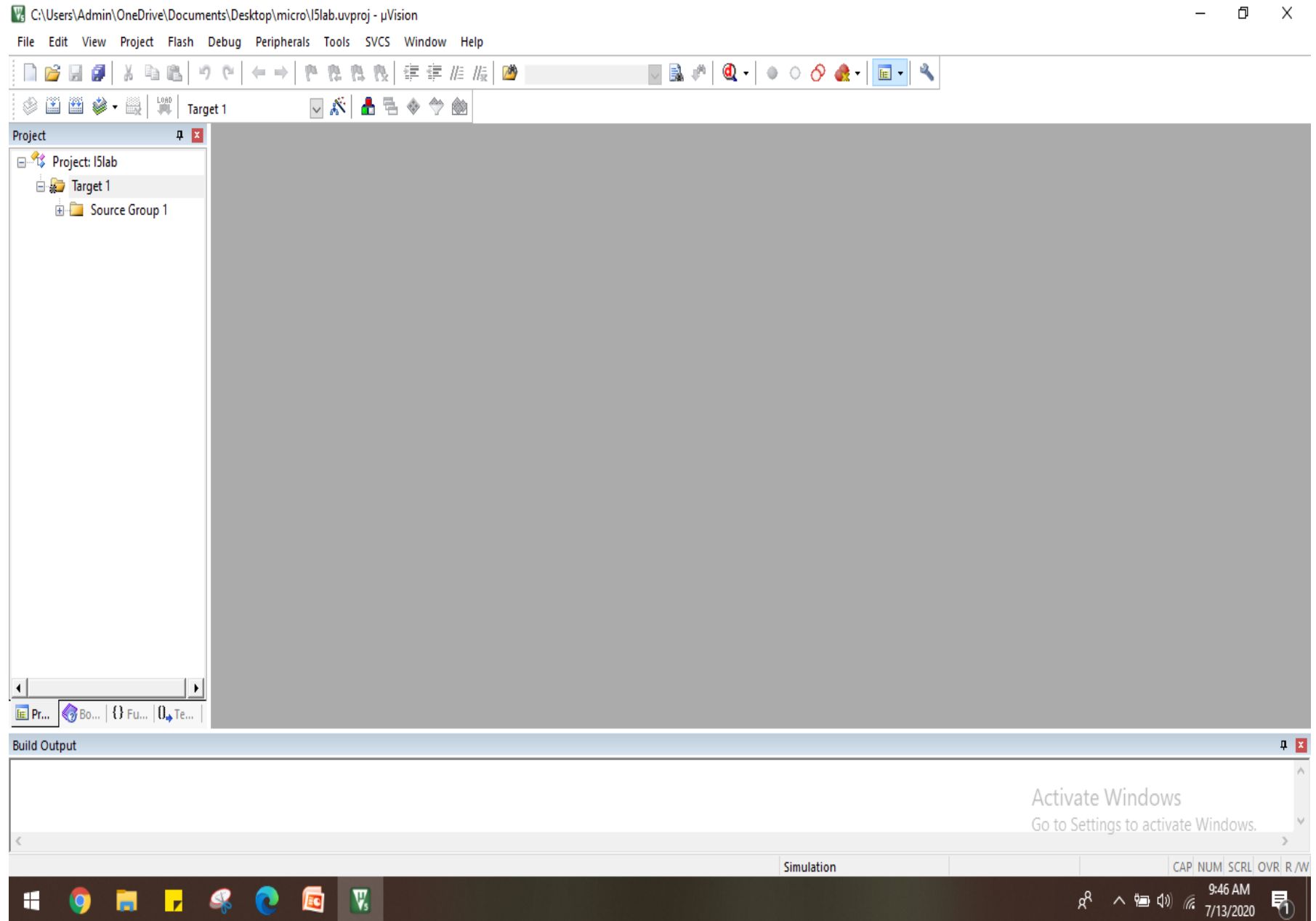
- Click Yes when asked whether to “Copy Standard 8051 Startup Code to Project Folder and Add File to Project.”



- Click Yes if asked “File <path>STARTUP.A51 already exists, overwrite?”

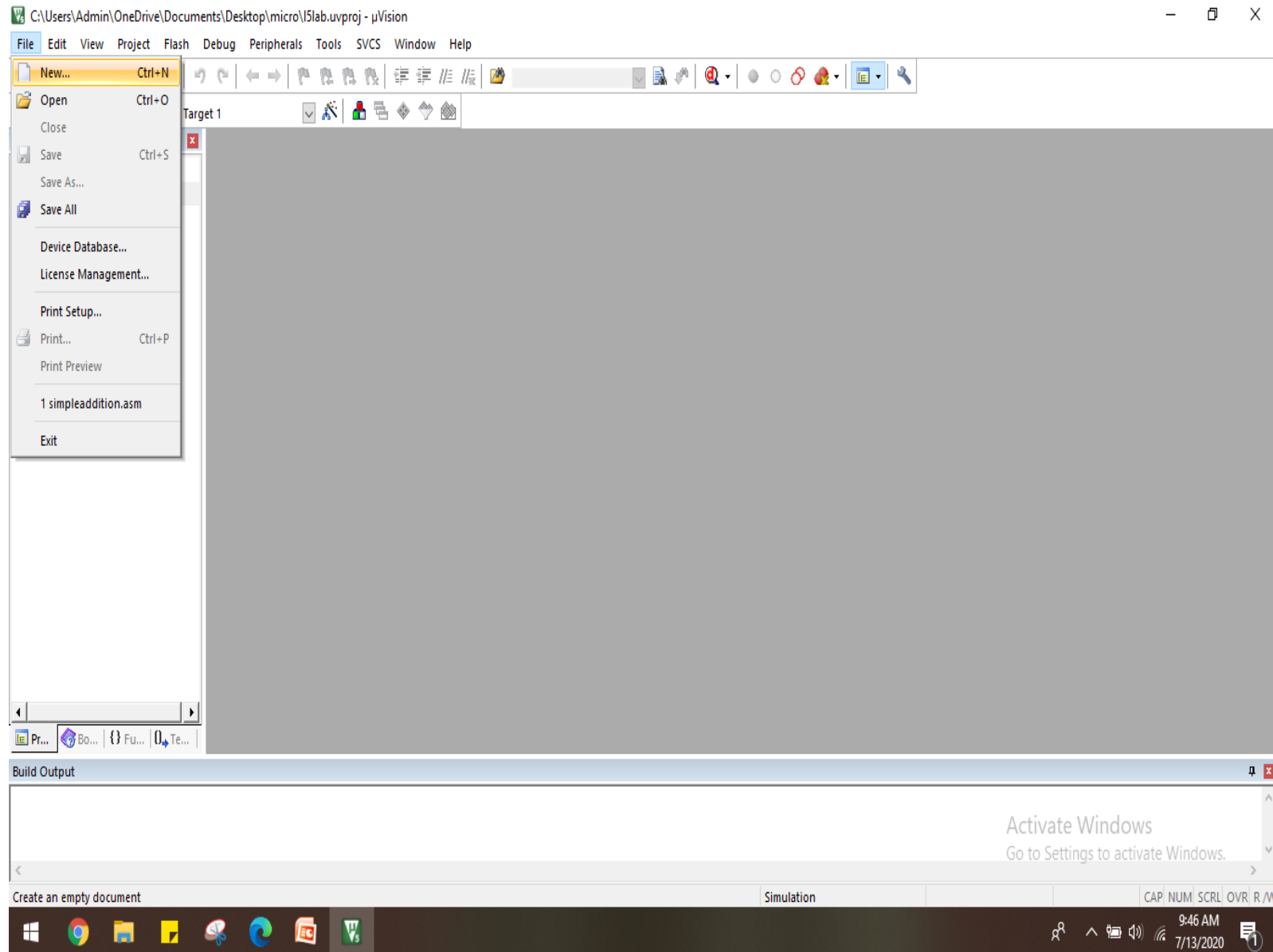


Blank space will appear



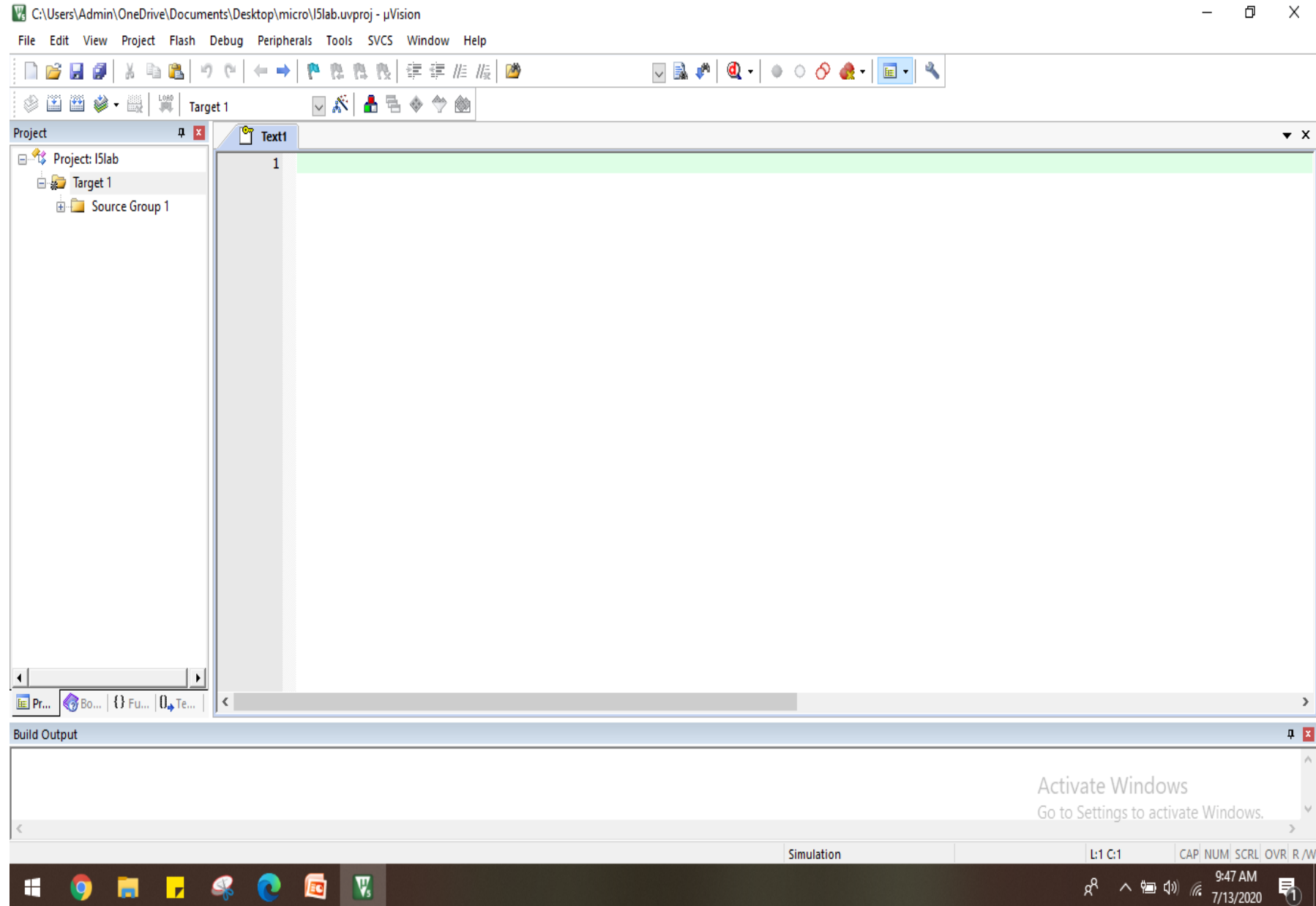
STEP 6: To create source file

Open a new file (File→New File)

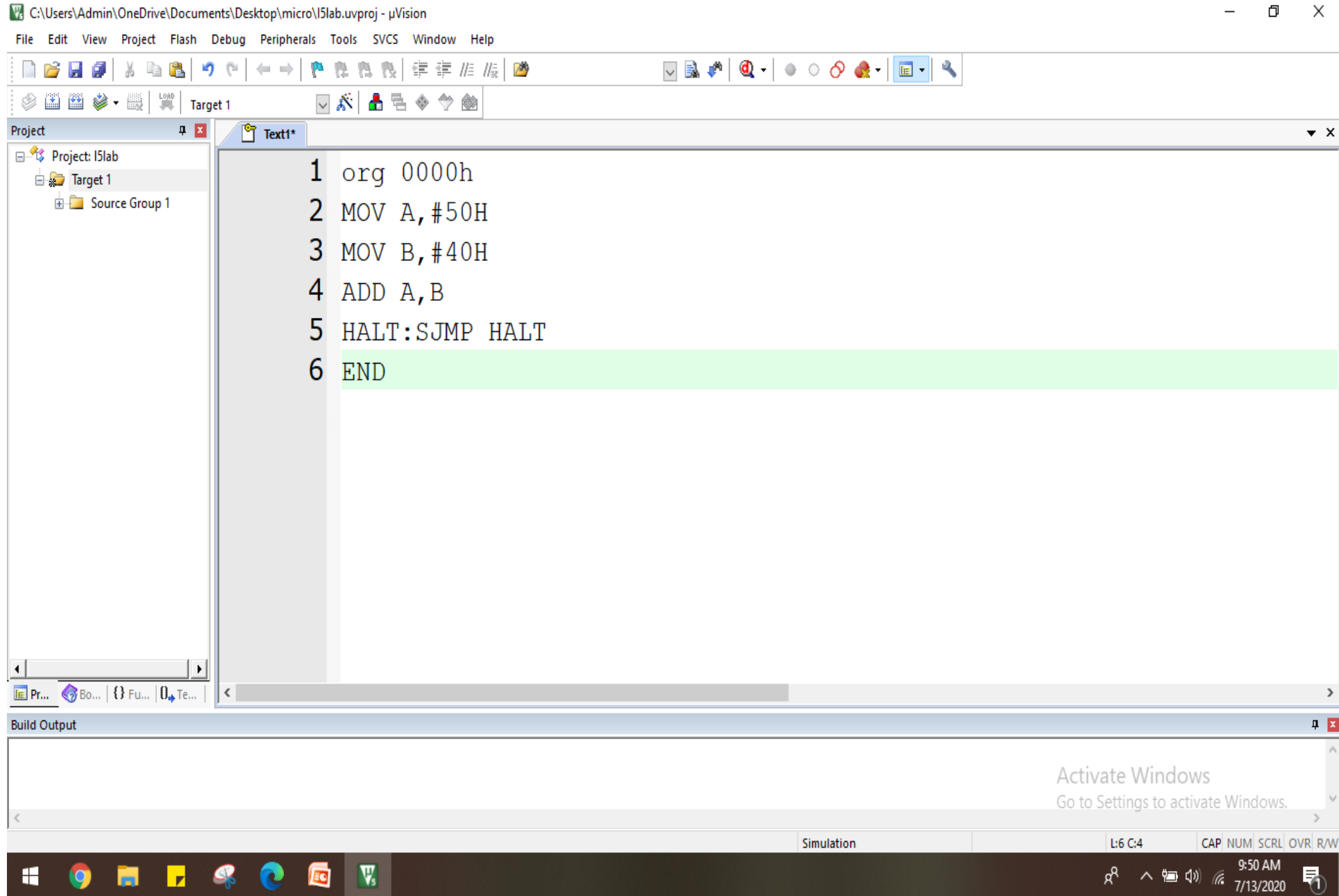


STEP 6 (Contd):

A new window will open with blank space to type the code



STEP 7: Type the code

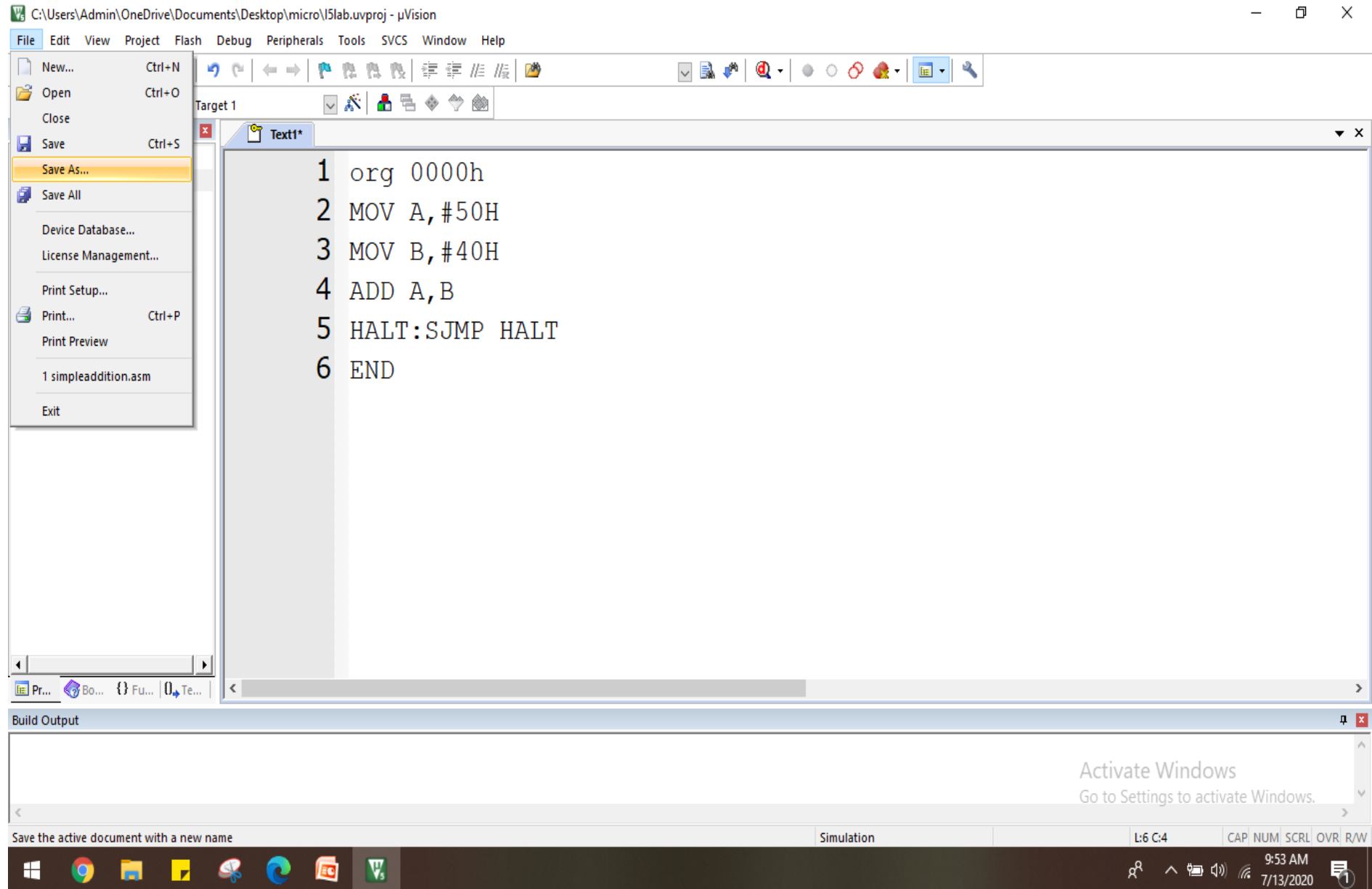


The screenshot shows the µVision IDE interface. The title bar indicates the file path: C:\Users\Admin\OneDrive\Documents\Desktop\micro\I5lab.uvproj - µVision. The menu bar includes File, Edit, View, Project, Flash, Debug, Peripherals, Tools, SVCS, Window, and Help. The toolbar contains various icons for file operations, editing, and simulation. The Project pane on the left shows a tree structure: Project: I5lab, Target 1, and Source Group 1. The main editor window, titled 'Text1*', contains the following assembly code:

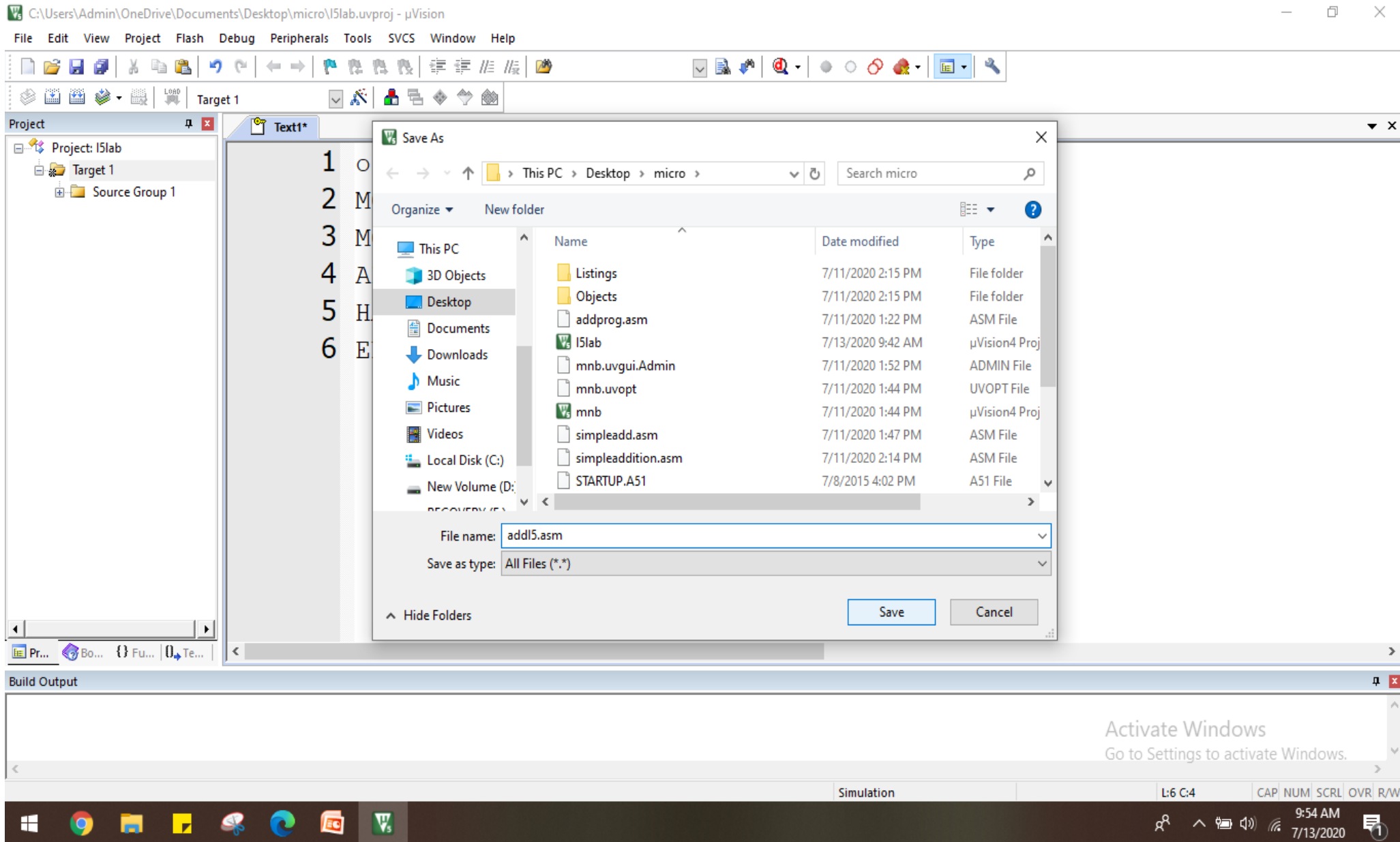
```
1 org 0000h
2 MOV A,#50H
3 MOV B,#40H
4 ADD A,B
5 HALT:SJMP HALT
6 END
```

The sixth line, '6 END', is highlighted in green. Below the editor is the 'Build Output' pane, which currently displays an 'Activate Windows' message: 'Activate Windows. Go to Settings to activate Windows.' The status bar at the bottom shows 'Simulation' mode, memory address 'L:6 C:4', and various registers: CAP, NUM, SCRL, OVR, R/W. The Windows taskbar at the very bottom shows the time as 9:50 AM on 7/13/2020.

STEP 8: Click file menu and save



STEP 9: Save the file name with extension as → .asm



The screenshot shows the µVision IDE interface. The 'Save As' dialog box is open, displaying the file explorer view. The file name is 'addI5.asm' and the save type is 'All Files (*.*)'. The background shows the µVision interface with a project named 'I5lab' and a target named 'Target 1'.

File name: addI5.asm
Save as type: All Files (*.*)

Buttons: Save, Cancel

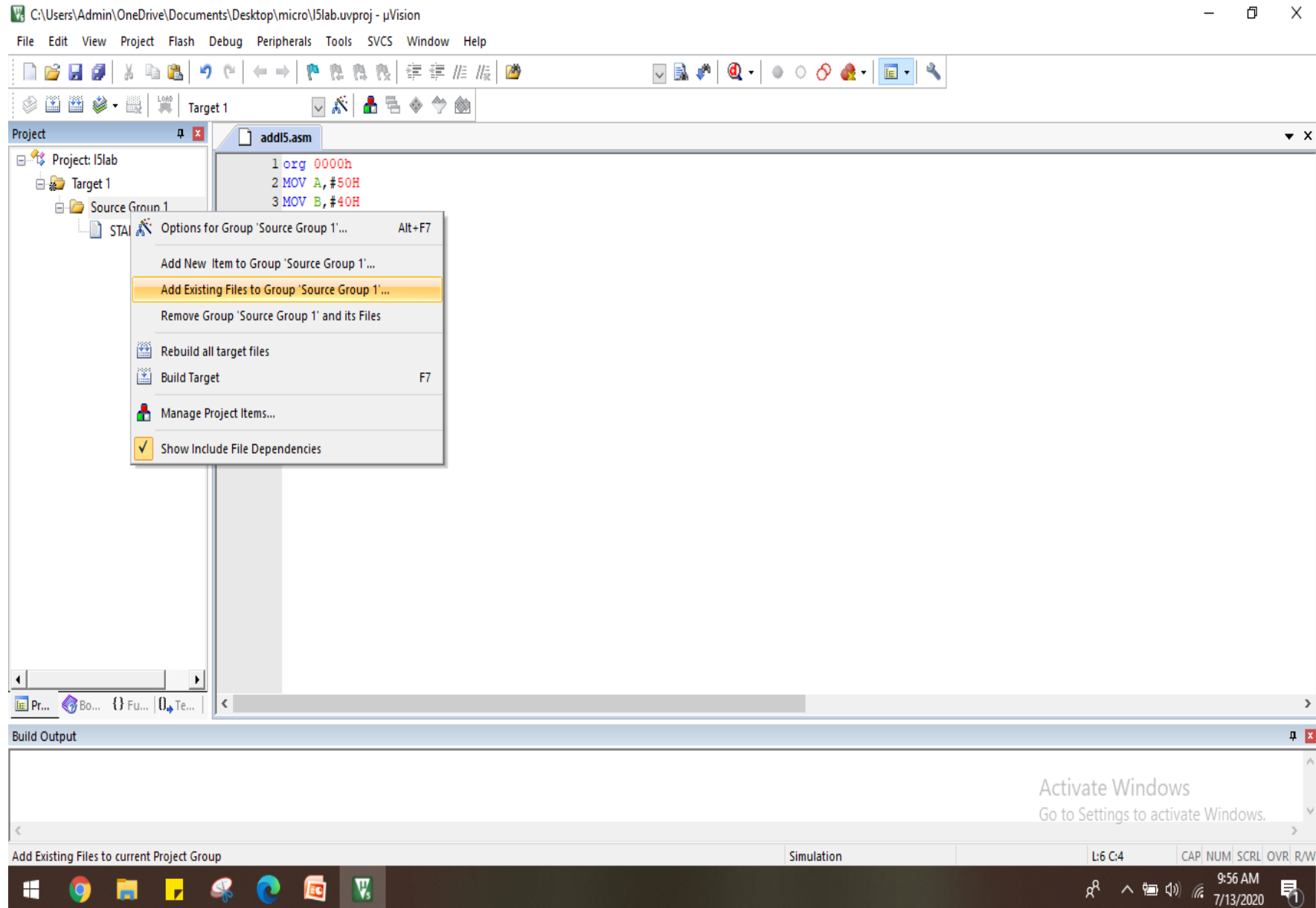
Build Output

Simulation

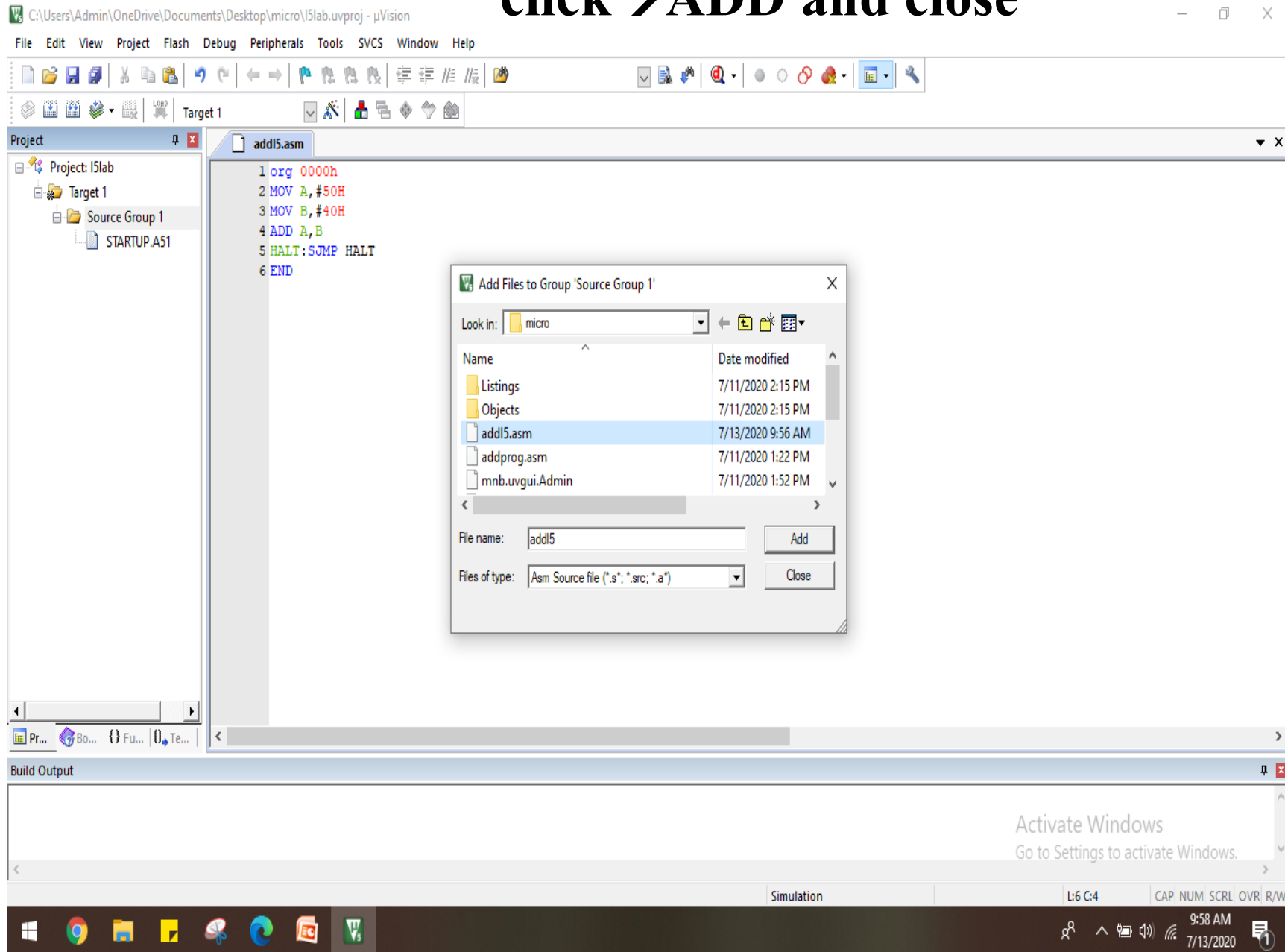
Activate Windows
Go to Settings to activate Windows.

9:54 AM
7/13/2020

Step 10: Right click the source group and select → Add existing file to the source group



Step 11: select the saved file from the folder and click→ADD and close



The screenshot shows the uVision IDE interface. The main window displays the assembly code for 'addl5.asm':

```

1 org 0000h
2 MOV A, #50H
3 MOV B, #40H
4 ADD A, B
5 HALT: SJMP HALT
6 END

```

The 'Project' pane on the left shows the project structure: Project: I5lab, Target 1, Source Group 1, and STARTUP.A51.

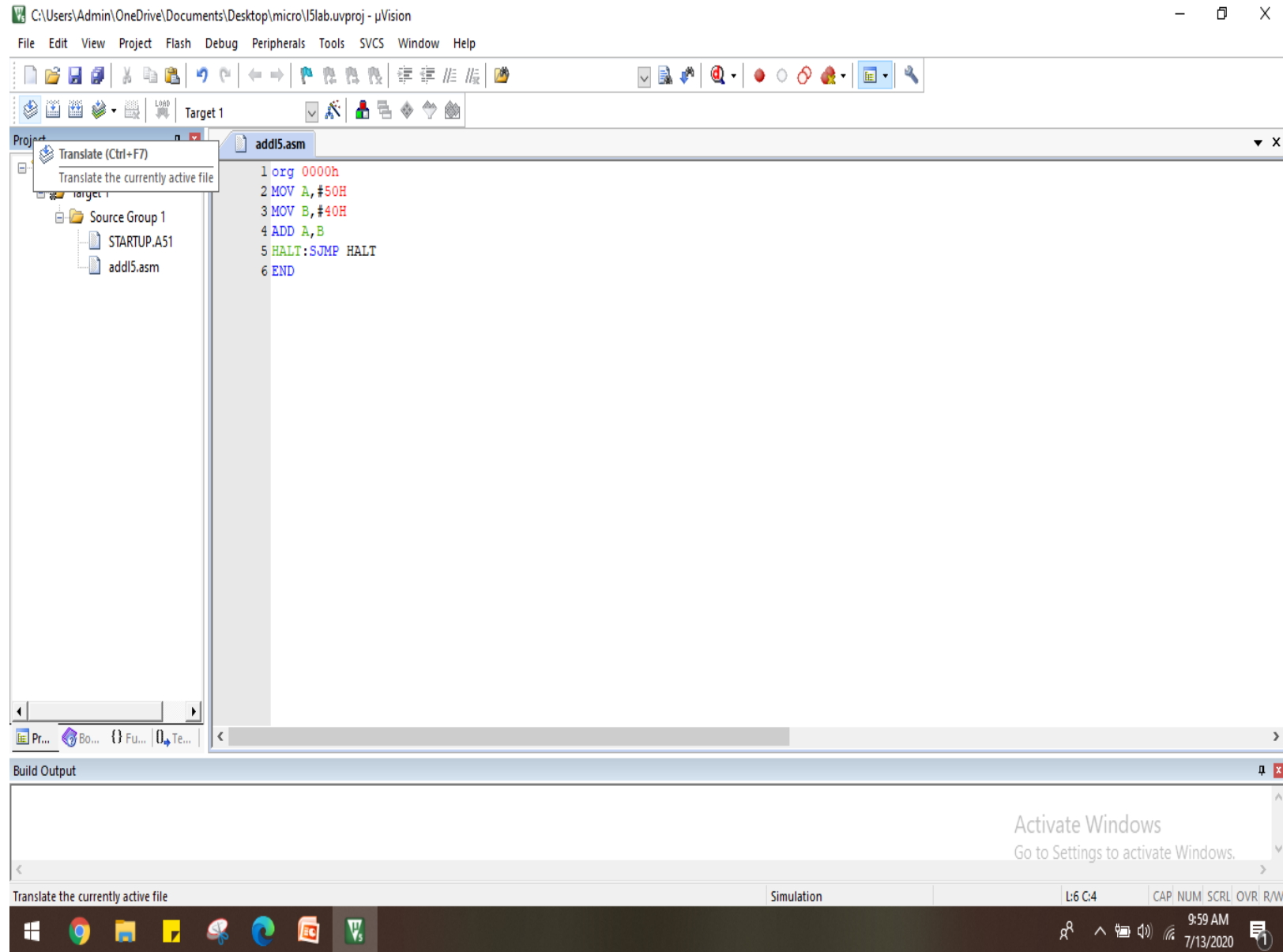
The 'Add Files to Group' dialog box is open, showing the 'micro' folder. The file list is as follows:

Name	Date modified
Listings	7/11/2020 2:15 PM
Objects	7/11/2020 2:15 PM
addl5.asm	7/13/2020 9:56 AM
addprog.asm	7/11/2020 1:22 PM
mnb.uvgui.Admin	7/11/2020 1:52 PM

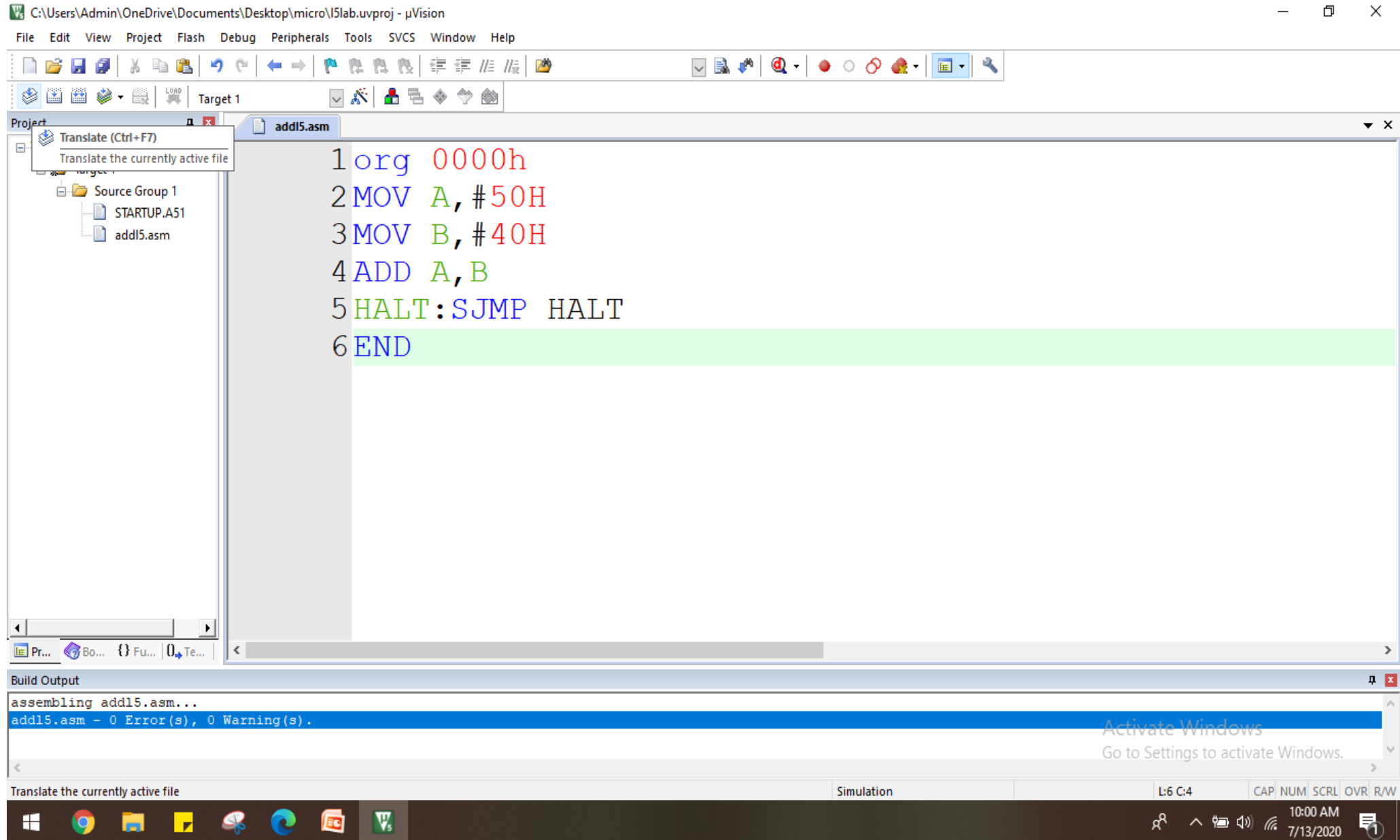
The 'File name' field contains 'addl5' and the 'Files of type' is set to 'Asm Source file (*.s; *.src; *.a)'. The 'Add' button is highlighted.

The 'Build Output' pane at the bottom shows the status 'Simulation' and 'L:6 C:4'.

Step12: Expand the source group in project work space and ensure your file is under source group



Step13: Select→ “Translate” on the left corner and check for svntax Errors



C:\Users\Admin\OneDrive\Documents\Desktop\micro\I5lab.uvproj - µVision

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Target 1

Project

- Source Group 1
 - STARTUP.A51
 - add15.asm

```
1 org 0000h
2 MOV A, #50H
3 MOV B, #40H
4 ADD A, B
5 HALT: SJMP HALT
6 END
```

Build Output

```
assembling add15.asm...
add15.asm - 0 Error(s), 0 Warning(s).
```

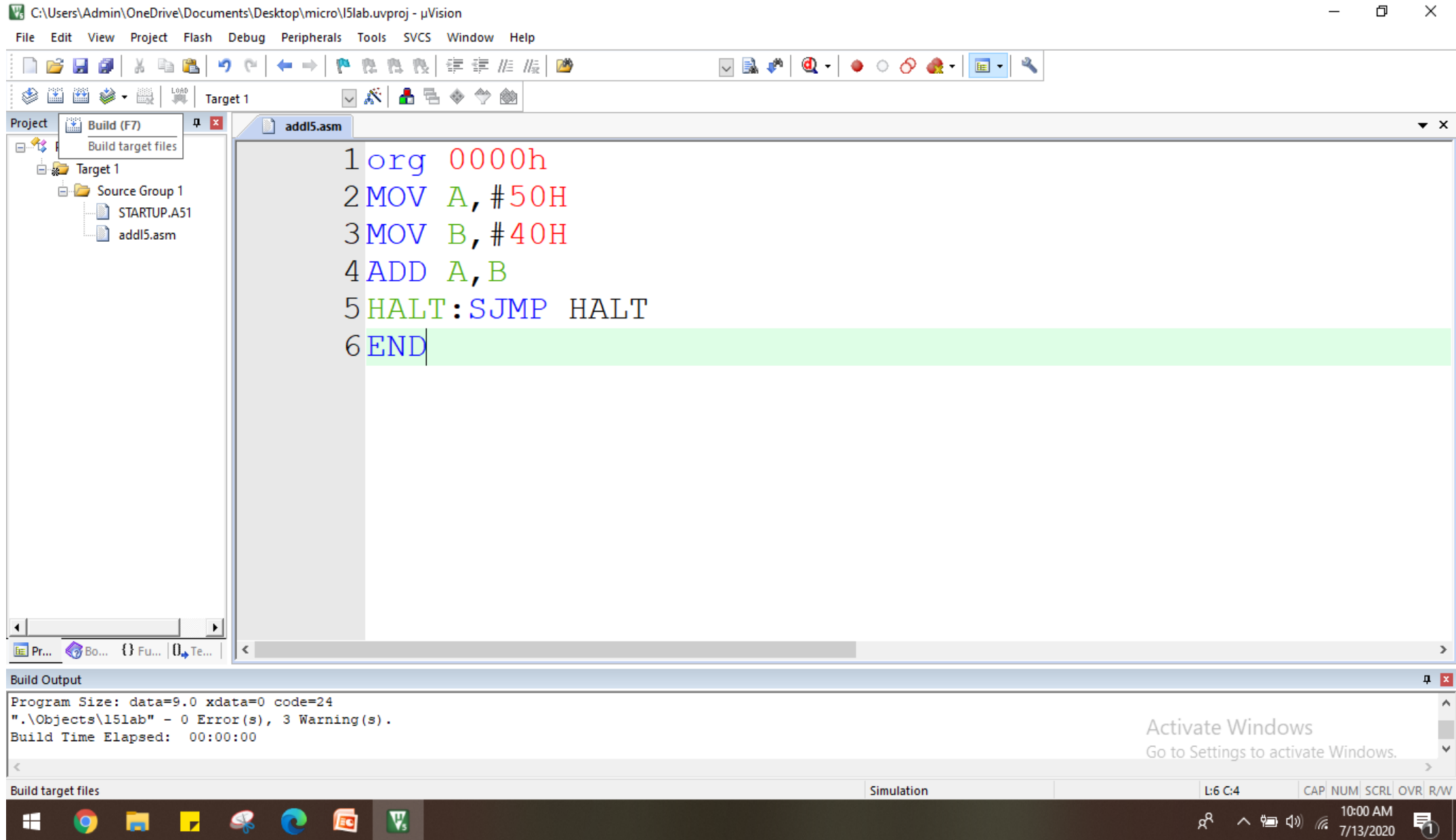
Translate the currently active file

Simulation

L:6 C:4 CAP NUM SCRL OVR R/W

10:00 AM 7/13/2020

Step14: Select → “Build” on the left corner



The screenshot shows the µVision IDE interface. The main window displays the assembly code for 'addI5.asm' with the following lines:

```
1 org 0000h
2 MOV A, #50H
3 MOV B, #40H
4 ADD A, B
5 HALT: SJMP HALT
6 END
```

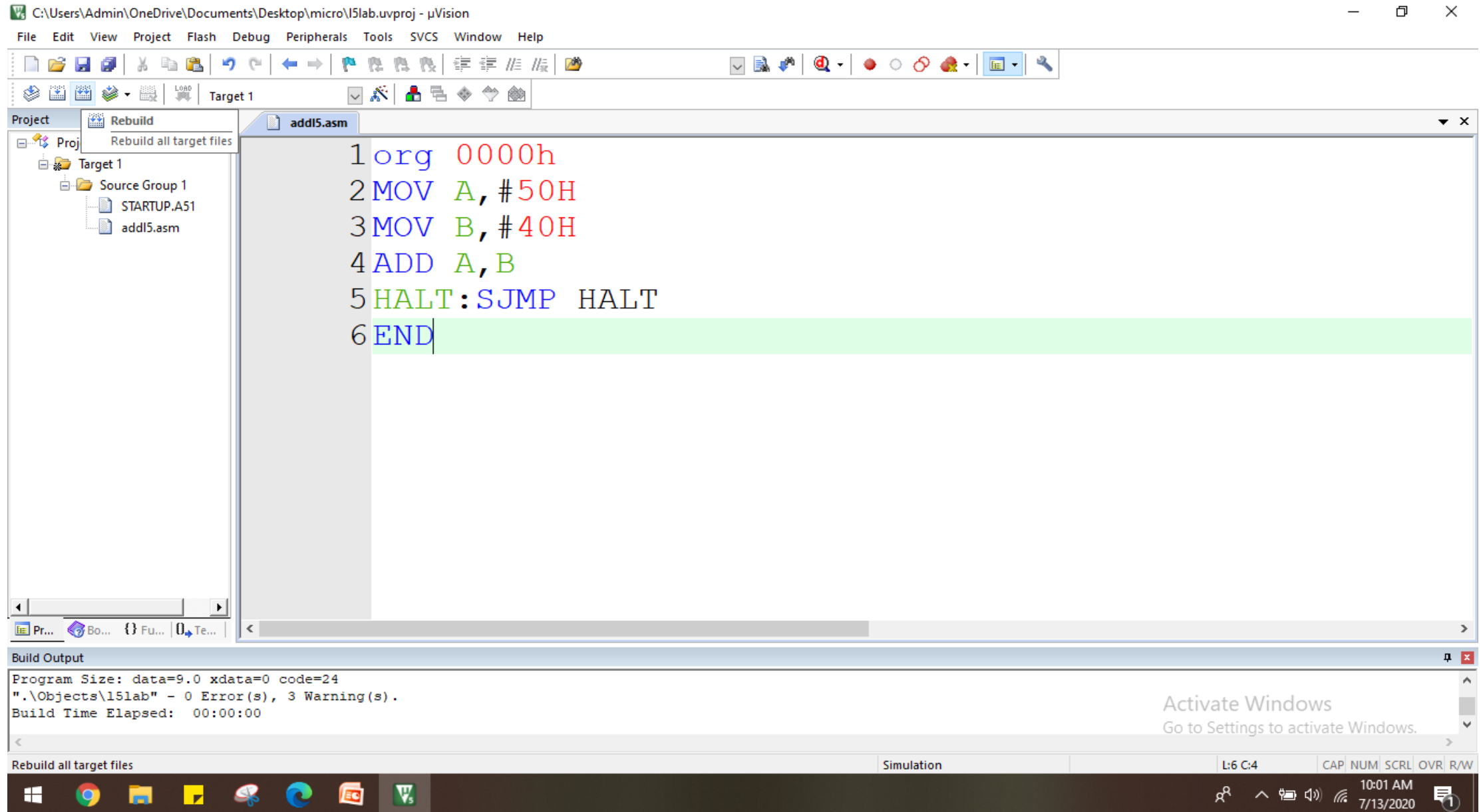
The left sidebar shows the Project tree with 'Target 1' selected. The 'Build' button (F7) is highlighted in the Project tree. The bottom status bar shows 'Build target files' and 'Simulation'.

The Build Output window at the bottom displays the following information:

```
Program Size: data=9.0 xdata=0 code=24
".\Objects\I5lab" - 0 Error(s), 3 Warning(s).
Build Time Elapsed: 00:00:00
```

An 'Activate Windows' watermark is visible in the bottom right corner of the Build Output window.

Step15: Select→ “Rebulid” on the left corner



The screenshot shows the µVision IDE interface. The title bar indicates the file path: C:\Users\Admin\OneDrive\Documents\Desktop\micro\l5lab.uvproj - µVision. The menu bar includes File, Edit, View, Project, Flash, Debug, Peripherals, Tools, SVCS, Window, and Help. The toolbar contains various icons for file operations, editing, and simulation. The Project window on the left shows a tree structure with 'Target 1' and 'Source Group 1' containing 'STARTUP.A51' and 'addl5.asm'. The 'Rebuild' button is highlighted in the Project window. The main editor displays the assembly code for 'addl5.asm':

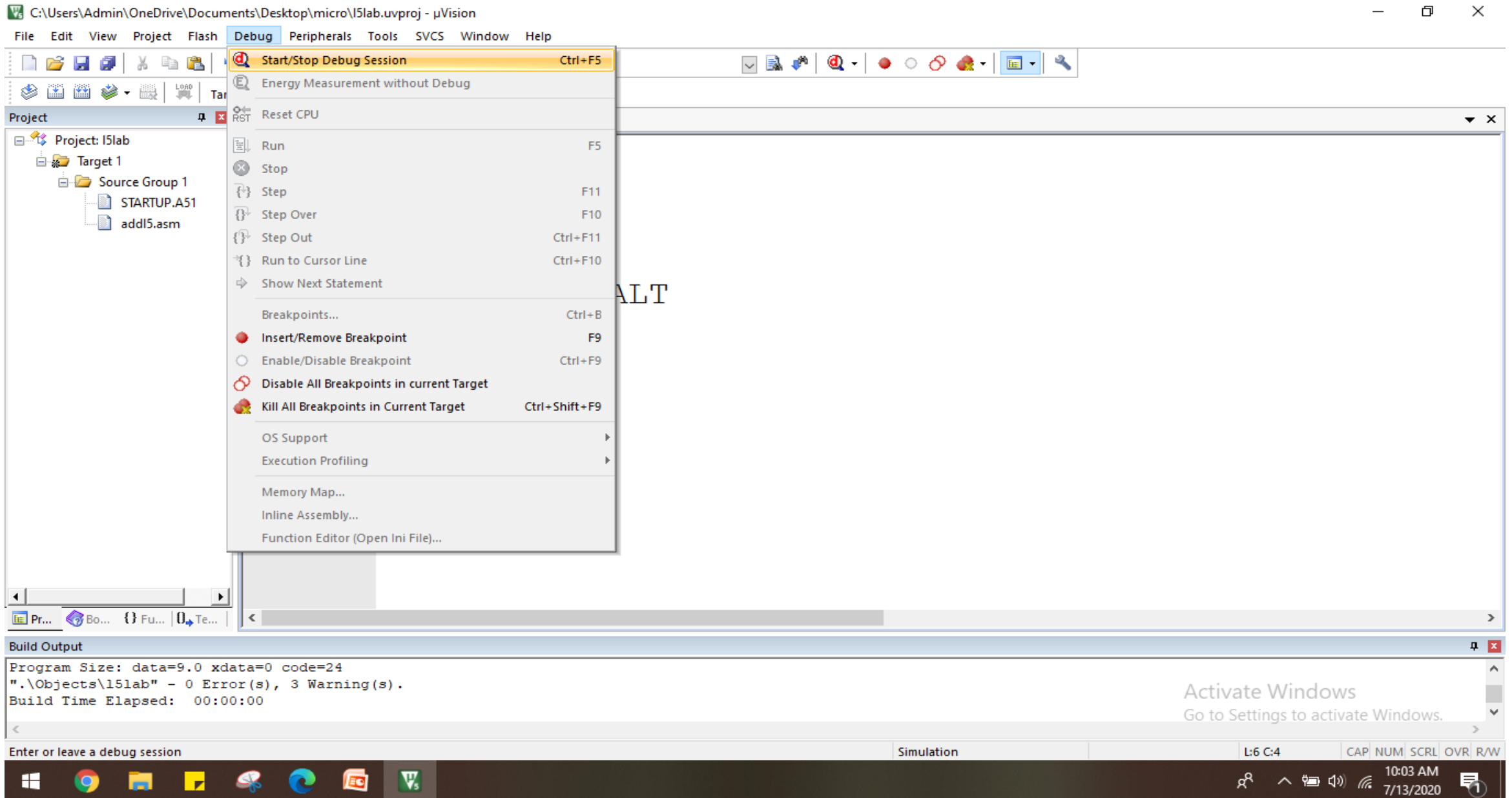
```
1 org 0000h
2 MOV A, #50H
3 MOV B, #40H
4 ADD A, B
5 HALT: SJMP HALT
6 END
```

The 'Build Output' window at the bottom shows the following text:

```
Program Size: data=9.0 xdata=0 code=24
".\Objects\l5lab" - 0 Error(s), 3 Warning(s).
Build Time Elapsed: 00:00:00
```

The status bar at the bottom indicates 'Rebuild all target files', 'Simulation', 'L:6 C:4', and 'CAP NUM SCRL OVR R/W'. The Windows taskbar at the bottom shows the time as 10:01 AM on 7/13/2020.

Step16: Debug→Start Debug



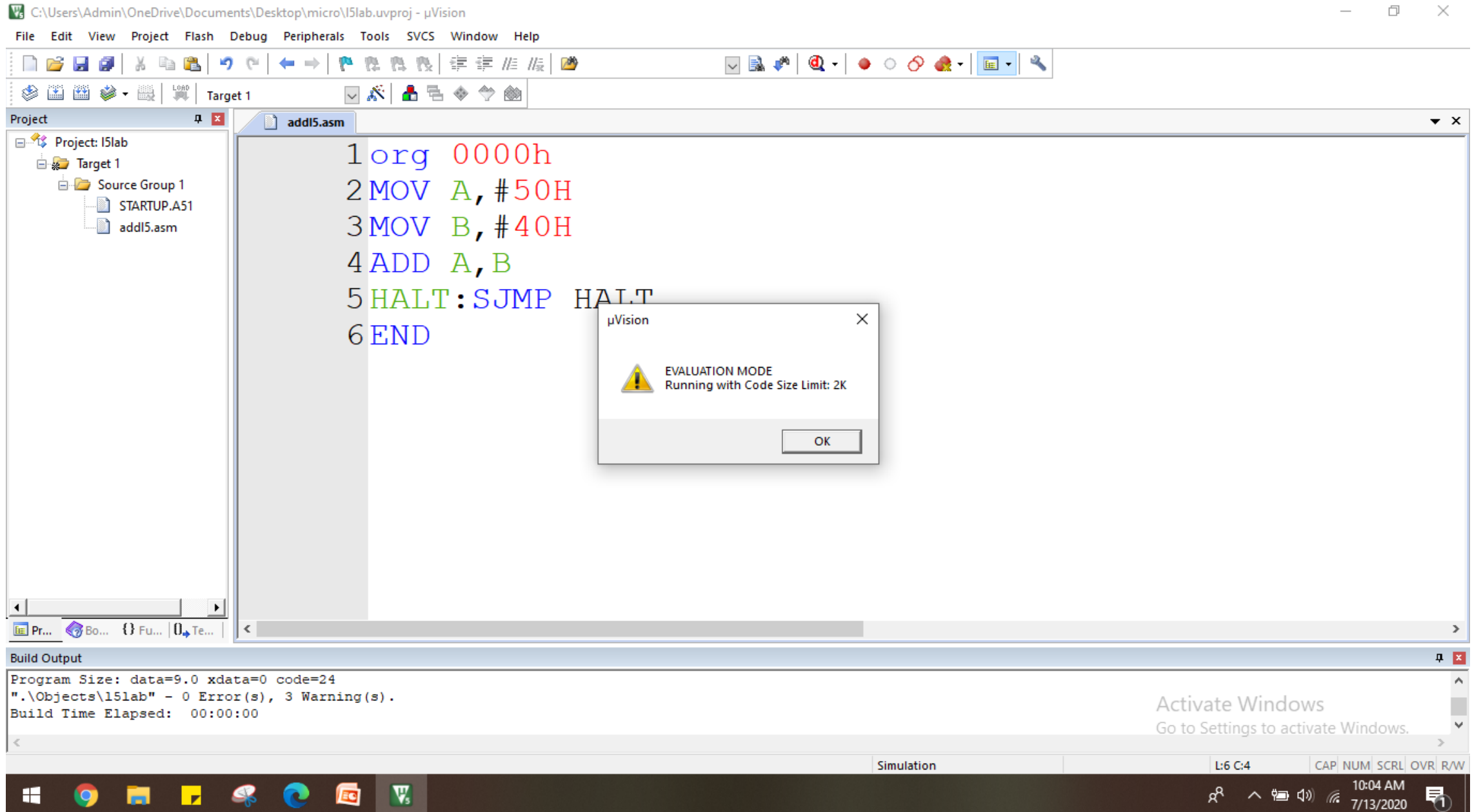
The screenshot shows the µVision IDE interface. The 'Debug' menu is open, and the 'Start/Stop Debug Session' option is highlighted. The menu includes various debugging actions such as 'Energy Measurement without Debug', 'Reset CPU', 'Run', 'Stop', 'Step', 'Step Over', 'Step Out', 'Run to Cursor Line', 'Show Next Statement', 'Breakpoints...', 'Insert/Remove Breakpoint', 'Enable/Disable Breakpoint', 'Disable All Breakpoints in current Target', 'Kill All Breakpoints in Current Target', 'OS Support', 'Execution Profiling', 'Memory Map...', 'Inline Assembly...', and 'Function Editor (Open Ini File)...'. The 'Project' pane on the left shows the project structure: 'Project: I5lab' containing 'Target 1' and 'Source Group 1' with files 'STARTUP.A51' and 'addI5.asm'. The 'Build Output' pane at the bottom shows the build results: 'Program Size: data=9.0 xdata=0 code=24', '".\Objects\I5lab" - 0 Error(s), 3 Warning(s).', and 'Build Time Elapsed: 00:00:00'. The status bar at the bottom indicates 'Enter or leave a debug session' and 'Simulation' mode.

ALT

Activate Windows
Go to Settings to activate Windows.

Enter or leave a debug session Simulation L:6 C:4 CAP NUM SCRL OVR R/W 10:03 AM 7/13/2020

Step17: Click →"Ok"



The screenshot shows the µVision IDE interface. The main window displays an assembly file named `addI5.asm` with the following code:

```
1 org 0000h
2 MOV A, #50H
3 MOV B, #40H
4 ADD A, B
5 HALT: SJMP HALT
6 END
```

A dialog box titled "µVision" is displayed in the center, showing a warning icon and the text:

EVALUATION MODE
Running with Code Size Limit: 2K

The dialog box has an "OK" button at the bottom right.

The Project Explorer on the left shows the project structure:

- Project: I5lab
 - Target 1
 - Source Group 1
 - STARTUP.A51
 - addI5.asm

The Build Output window at the bottom shows the following information:

```
Program Size: data=9.0 xdata=0 code=24
".\Objects\I5lab" - 0 Error(s), 3 Warning(s).
Build Time Elapsed: 00:00:00
```

The status bar at the bottom indicates the current mode is "Simulation" and the code size is "L:6 C:4". The Windows taskbar at the very bottom shows the time as 10:04 AM on 7/13/2020.

Registers will appear on the left

C:\Users\Admin\OneDrive\Documents\Desktop\micro\l5lab.uvproj - µVision

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
Regs	
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
Sys	
a	0x00
b	0x00
sp	0x07
sp_max	0x07
dptr	0x0000
PC	0x0000
states	0
sec	0.00000000
psw	0x00

Disassembly

```

C:0x0000 7450 MOV A, #0x50
3: MOV B, #40H
C:0x0002 75F040 MOV B(0xF0), #0x40
4: ADD A, B
  
```

addl5.asm

```

1 org 0000h
2 MOV A, #50H
3 MOV B, #40H
4 ADD A, B
5 HALT: SJMP HALT
6 END
  
```

Command

Running with Code Size Limit: 2K
Load "C:\\Users\\Admin\\OneDrive\\Documents\\Desktop\\micro\\Objects\\l5lab"

ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess

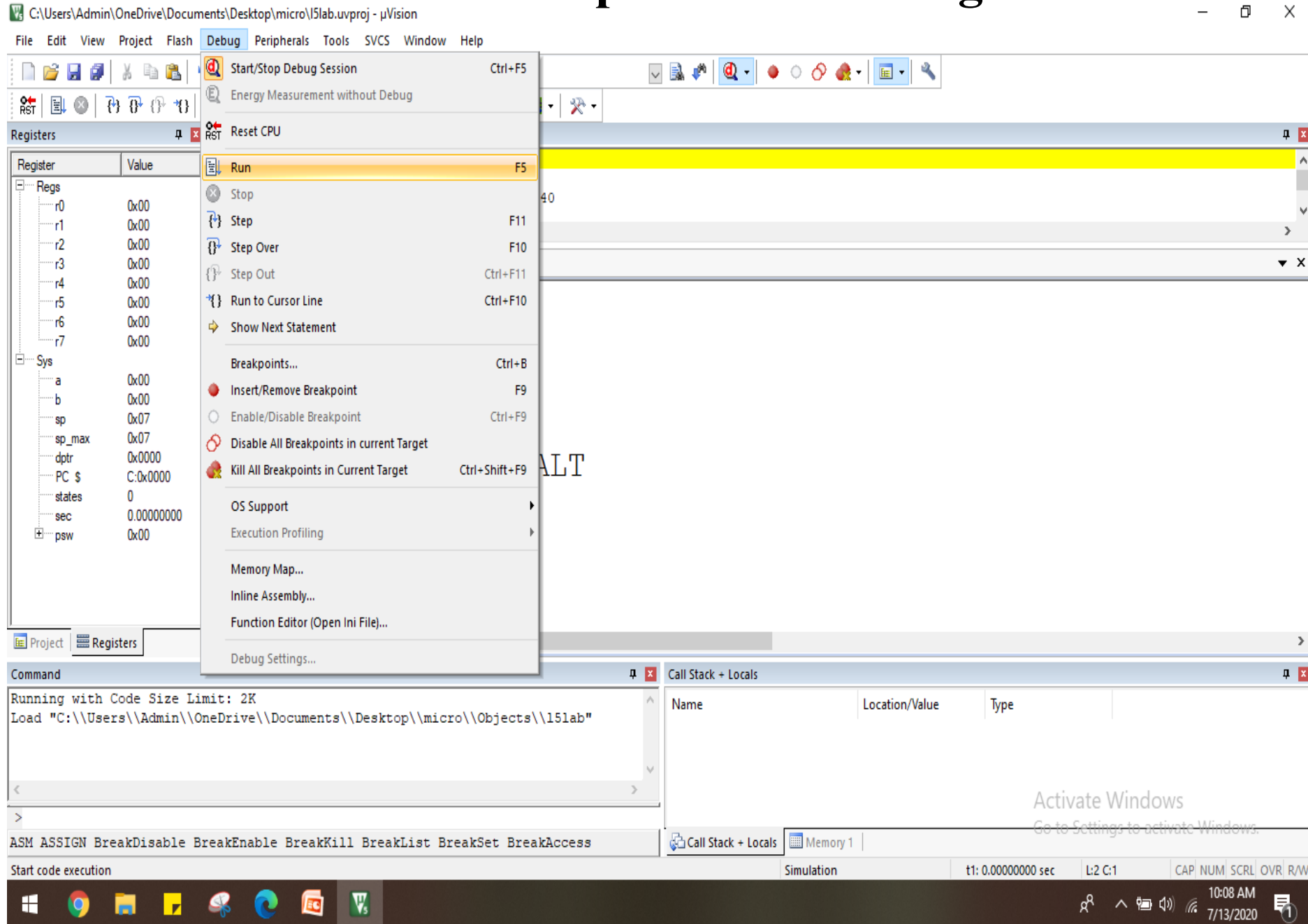
Call Stack + Locals

Name	Location/Value	Type
Activate Windows Go to Settings to activate Windows.		

Simulation t1: 0.00000000 sec L:2 C:1 CAP NUM SCRL OVR R/W

26

Step18: Click Debug→Run



The screenshot shows the Keil uVision IDE interface. The 'Debug' menu is open, and the 'Run' option (F5) is highlighted. The 'Registers' window on the left shows the status of various registers. The 'Command' window at the bottom displays the execution status.

Registers Window:

Register	Value
Regs	
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
Sys	
a	0x00
b	0x00
sp	0x07
sp_max	0x07
dptr	0x0000
PC	0x0000
states	0
sec	0.00000000
psw	0x00

Command Window:

```
Running with Code Size Limit: 2K
Load "C:\Users\Admin\OneDrive\Documents\Desktop\micro\Objects\151lab"
```

Debug Menu Options:

- Start/Stop Debug Session (Ctrl+F5)
- Energy Measurement without Debug
- Reset CPU
- Run (F5)
- Stop
- Step (F11)
- Step Over (F10)
- Step Out (Ctrl+F11)
- Run to Cursor Line (Ctrl+F10)
- Show Next Statement
- Breakpoints... (Ctrl+B)
- Insert/Remove Breakpoint (F9)
- Enable/Disable Breakpoint (Ctrl+F9)
- Disable All Breakpoints in current Target
- Kill All Breakpoints in Current Target (Ctrl+Shift+F9)
- OS Support
- Execution Profiling
- Memory Map...
- Inline Assembly...
- Function Editor (Open Ini File...)
- Debug Settings...

Call Stack + Locals Window:

Name	Location/Value	Type

Simulation Status:

Start code execution | Simulation | t1: 0.00000000 sec | L2 C:1 | CAP NUM SCRL OVR/ R/W

Debugging in Keil is a crucial part of the embedded systems development process, enabling developers to ensure the correctness, reliability, and performance of their software on the target hardware.

Check the results

C:\Users\Admin\OneDrive\Documents\Desktop\micro\l5lab.uvproj - uVision

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
Sys	
a	0x90
b	0x40
sp	0x07
sp_...	0x07
dptr	0x0000
PC \$	C:0x0007
states	921119490
sec	276.33587463
psw	0x04
p	0
f1	0
o..	1
rs	0
f0	0
a..	0
cy	0

Disassembly

```

5: HALT:SJMP HALT
C:0x0007 80FE SJMP HALT (C:0007)
C:0x0009 00 NOP
C:0x000A 00 NOP
  
```

addl5.asm

```

1 org 0000h
2 MOV A, #50H
3 MOV B, #40H
4 ADD A, B
5 HALT:SJMP HALT
6 END
  
```

Command

Running with Code Size Limit: 2K
Load "C:\\Users\\Admin\\OneDrive\\Documents\\Desktop\\micro\\Objects\\l5lab"

Call Stack + Locals

Name	Location/Value	Type
ADDL5	C:0x0007	
B	0x40 '@'	uchar

Simulation

t1: 276.33587463 sec L:5 C:1 CAP NUM SCRL OVR R/W

10:12 AM 7/13/2020

Any Questions?



Thank you!

